PassFrame tuition

Angles & 2D Shapes

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. Acute

2. Obtuse

3. C — 210°

4. Reflex

5. 180°

6. 360°

7. 5

8. 6

9. A and D (a triangle's interior angles always sum to 180°, and this is true for every triangle regardless of shape)

10. 8

11. 180°

12. Equilateral

13. Isosceles

14. 4

15. Straight angle

Standard

1. 108°

2. 155°

3. B — 65°

4. 85°

5. 90° each

6. 70°

7. 48°

8. A, B and D (vertically opposite angles are equal, formed where two lines cross, and there are two such pairs)

9. 70° and 40°

10. 55°

11. 90°

12. 72° and 108°

13. 105°

14. 50°

15. 120°

Standard / Hard

1. 40°, 60° and 80°

2. 45°, 135°, 45° and 135°

3. A — x=36, largest angle = 144°

4. 108°

5. 9 sides

6. A, C and D (exterior angles of any polygon sum to 360°, each pairs with its interior angle to make 180°, and a regular polygon's exterior angles are all equal)

7. x = 36; angles are 82° and 98°

8. 80° each

9. 12 sides

10. b = 180°, c = 90°

11. Interior at that vertex = 50°; third angle = 95°

12. 65°, 115° and 115°

13. A = 80°, B = 40°, C = 60°

14. 10 sides; each angle = 144°

15. 108°

Homework — Answers

Easy / Standard

1. Acute

6. 360°

11. Scalene

2. Obtuse

3. C — right angle

4. Reflex

5. 180°

7. 7

8. 9

9. A and C (a quadrilateral's interior angles always sum to 360°, whatever its shape)

10. 10

12. Obtuse-angled triangle

13. 6

14. Reflex angle

15. 3

Standard

1. 96°

2. 140°

3. B — 60°

4. 80°

5. 180° each

6. 84°

7. 63°

8. A, B, D and E (an isosceles triangle has two equal sides, two equal base angles, and can also happen to be right-angled — it does not need all angles equal, which would make it equilateral)

9. 55° and 70°

10. 62°

11. 105° each

12. 54° and 126°

13. 85°

14. 65°

15. 135°

Standard / Advanced

1. 45° (since 3b + b = 180)

2. B — 15 (360 ÷ 24)

3. 30° (angles are 100°, 50°, 30°)

4. A, C and E (co-interior angles add to 180°, lie on the same side of the transversal, and only exist between a pair of parallel lines)

5. 90° each — a rectangle (or square)

6. x = 50

7. 9 slices

8. 40° each

9. 25°

10. 83°

11. 9 sides

12. 55°; a right-angled (scalene) triangle

13. 65°, 115° and 65°

14. 80°

15. 5 sides